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Pricing and return strategy of online retailers based on return insurance

Minglun Ren [a], Jiqiong Liu [a,b,c,*], Shuai Feng [a,b,c], Aifeng Yang [a]

[a] School of Management, Hefei University of Technology, Hefei, 230009, China
[b] School of Business, Fuyang Normal University, Fuyang, 236037, China
[c] Anhui Provincial Key Laboratory of Regional Logistics Planning and Modern Logistics Engineering, Fuyang, 236037, China

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ABSTRACT

With the development of e-commerce, online shopping has become increasingly common, and as a result, consumers inevitably encounter the problem of returns. Therefore, pricing, return policy and return insurance strategy have attracted considerable research attention. In this paper, we construct four models to study pricing, return policy and return insurance strategy. We show that when a product's net residual value is greater than or equal to zero, online retailers should offer a money-back guarantee (MBG) return policy; however, they do not have to offer free return insurance because the latter does not increase their market share and profit, nor does it increase consumer surplus. The optimal strategy of insurance providers is unaffected by whether the policyholder is an online retailer or a consumer and should be neutral, which helps insurance providers gain the trust of policyholders. Consumers should buy products only when online retailers offer an MBG return policy; however, consumers should not do so if the online retailer provides free return insurance.

1. Introduction

The rapid development of e-commerce has given rise to a growing online shopping trend. However, due to the lack of "touch and experience" when shopping online, consumers often cannot accurately assess the true value of products. Therefore, customers often return their purchases, and, as the scale of online shopping increases, the problems associated with customer returns are becoming increasingly serious. According to different product categories, the average return rate for online purchases is approximately 15–40%, which is three to four times the 5–10% return rate for physical stores. The return rate for clothing and shoes purchased online is usually 30–40% (Reagan 2019), and the return rate for fashion and luxury products is even higher (Son et al., 2019). Given the frequency of customer returns, some online retailers focus on creating a good return experience as a means of improving consumer service, and they provide a relaxed and customer-friendly return policy such as a money-back guarantee (MBG). An MBG is popular with consumers because they can return unsatisfactory products and obtain a full refund (Akçay et al., 2013; Heydari et al., 2017). However, processing customer returns is very expensive for online retailers. Loss from online returns in the United States in 2019 is estimated to account for 9.6% of total retail online sales, causing online retailers to lose 41 billion dollars in sales (Appriss 2020). Therefore, to reduce and limit the costs of processing customer returns, some online retailers choose to tighten their free return policies (Su 2009), with some not even providing any return options (Hsiao and Chen 2015). Because frequent customer returns may lead to a large number of returned products that must be processed and high associated costs, return policies decisively affect the profitability of retailers (Ülkü et al., 2013; Li et al., 2018).

Offering return services is an important strategy for retailers to maintain market competitiveness (Chen and Chen 2017a). From the perspective of consumers, return services reduce the purchase risks and losses caused by actual commodity mismatches; from the perspective of online retailers, return services increase consumer willingness to purchase (Suwelack et al., 2011). However, while the return service strategy has boosted consumer satisfaction and spurred sales, some online retailers are deterred from adopting this strategy, given the growing problems with customer returns. Tobin Moore, the CEO of Optoro pointed out that in the next several years, with the development of global e-commerce, the amount of returns is going to be over a trillion dollars per year (Reagan 2019). The cost of product returns is also related to the hidden costs of the return process (Ofek et al., 2011), which includes not only the costs of repackaging, refurbishment processing and inventory clearance but also the cost of reverse logistics. Moreover, consumer returns of online products are so frequent that

* Corresponding author. School of Management, Hefei University of Technology, No.193, Tunxi Road, Baohe District, Hefei, 230009, China.
E-mail addresses: renml@hfut.edu.cn (M. Ren), liujiqiong@mail.hfut.edu.cn (J. Liu), fengshuai@mail.hfut.edu.cn (S. Feng), yangaifeng@hfut.edu.cn (A. Yang).

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processing them requires a great deal of extra reverse logistics costs (Shulman et al., 2010). However, it is unclear who should pay for the reverse logistics costs of customer returns, which is a burden to both parties involved in the transaction. Furthermore, as e-commerce grows globally, just as consumers increasingly expect free shipping for online purchases, they also increasingly expect free returns.

To resolve this contradiction, on November 9, 2010, China's Huatai Insurance Group, in conjunction with the Taobao platform, launched the first true Internet insurance product in China – "online shopping return freight loss insurance" (referred to as "return insurance"). Return insurance is a type of online insurance that is fully combined with online shopping, which can help online retailers provide more relaxed return policies and help consumers reduce return losses. The return insurance jointly launched by the Huatai Insurance Group and Taobao platform includes the seller's version and the buyer's version. Under the seller's return insurance, the online retailer provides the return service with free return insurance for the consumer, and the consumer is compensated for the reverse freight paid by the insurance company after the return, which can increase the consumer's likelihood of purchasing and willingness to pay. Consumers can also purchase buyer's return insurance when the online retailer provides the return service but does not offer free return insurance. In this case, the reverse freight compensation paid by the insurance company can be obtained only after the consumer purchases buyer's return insurance.

Researchers have turned their focus to whether return insurance can improve the profitability of online retailers and enhance consumer willingness to pay and consumer surplus. The question of what type of pricing and return strategies to adopt is also a key issue for online retailers to consider. In this paper, we study the pricing decision and return strategy selection of an online retailer in the presence of return insurance and the return insurance strategy of an insurance provider. Specifically, we consider the following issues:

(1) When should online retailers choose to offer return services, and what type of return strategy can increase the market share and profitability of online retailers?
(2) How do consumer returns, the return insurance premium and the return freight compensation affect the performance of online retailers and consumer surplus?
(3) Should online retailers offer consumers free return insurance? If not, should consumers purchase return insurance themselves? Is there a difference with respect to who purchases return insurance?
(4) Does the performance of the insurance provider relate to who the policyholder is?

To address these questions, we consider a sales system composed of an online retailer and an insurance provider. We use a classical maximization microeconomic consumer model to examine how the retailer determines customer return strategies (whether to provide an MBG return policy) and the corresponding pricing strategies. How do the different strategies affect the return insurance strategy of the insurance provider?

We show that when the net residual value of a returned product is positive, the online retailer should provide the MBG return policy, and we argue that the online retailer should strive to improve the net residual value of the returned product in the online channel by reducing the loss from processing consumer returns because improving the net residual value of the returned product and making it positive can help online retailers by allowing them to benefit from implementing the MBG return strategy. In addition, the online retailer can maximize its profit when offering the MBG return policy without offering return insurance to customers.

Furthermore, the highest consumer surplus is obtained when the online retailer provides the MBG return policy but does not offer return insurance, which means that consumers do not have to consider whether to obtain return insurance from retailers for free or purchase return insurance themselves when purchasing a certain product. This is because the consumer obtains the benefit but the price has been paid. In other words, return insurance causes the cost of consumer purchases to rise.

Finally, we find that the insurance provider is neutral, and the optimal pricing of return insurance does not change with respect to whether the policyholder is an online retailer or a consumer. This also helps the insurance provider gain the trust of the policyholders.

The remainder of this paper is organized as follows. Section 2 provides a literature review. Section 3 presents a description of the problem and the modeling of four situations. Section 4 discusses the return policy of the online retailer and its decision of whether to offer return insurance. Section 5 concludes with a discussion of managerial insights and recommendations for future research.

2. Literature review

Two strands of literature are relevant to our research. The first concerns return motivation and return policies, and the second examines Internet return insurance.

2.1. The research on return motivation and return policies

2.1.1. Return motivation

Consumers' motives for returning products are mostly due to issues of dissatisfaction including product failures, product damage, wrong delivery, and delays in transportation (Cole 1989; Chu et al., 1998; King and Dennis 2006; Hong and Pavlou 2014; Son et al., 2019). However, it is interesting that some studies assume that product returns are not related to product dissatisfaction or failure but are caused by consumers' impulsive consumption and hedonistic attitudes (Lee 2014). Moreover, a research report (Appriss 2020) showed that part of the return motivation is fraudulent return (for example, the purchased goods are genuine; however, the returned goods are fakes). In 2019, fraudulent returns in the United States accounted for 8.8% of the total return value, causing retailers to lose 27 billion dollars.

Furthermore, manufacturers provide returns to retailers although limited to excess inventory (e.g., Padmanabhan and Png 1997; Emmons and Gilbert 1998; Iyer and Villas-Boas 2003; Padmanabhan and Png 2004; Wang 2004; Pasternack 2008; Bandyopadhyay and Paul 2010; Cachon and Kök 2010). As seen from the above literature, the manufacturer's return policy is a guarantee for the retailer, which faces uncertain demand, thereby eliminating or reducing the retailer's sales risk and encouraging the retailer to maintain more inventory. This incentive is similar to the effect of a retailer's return policy on stimulating consumer purchases. In addition, some manufacturers only provide consumers with returns in their online store channels; the motivation is to use this as a competitive strategy to sell products directly to end consumers (Alaei et al., 2020).

2.1.2. Return policies

As a type of after-sales service, the return policy is a decisive factor for consumers to purchase products (Chen and Chen 2015). Regarding the research on return policies, most researchers believe that a loose return policy (such as MBG) can reduce consumers' perception of risk and stimulate purchase emotional response, thereby enhancing customers' willingness to purchase and enhance their final willingness to pay (Suwelack et al., 2011; Chen and Chen 2015; Jeng 2017; Oghazi et al., 2018). This is because a loose return policy as a market signal mechanism can promote trust between buyers and sellers, thereby enhancing consumers' purchase willingness (Oghazi et al., 2018) and ensuring that retailers gain a greater market share in a competitive market (Chen and Chen 2015). The generosity of a return policy tends to increase the consumer's purchase willingness by increasing the perceived value of the return policy; in a competitive environment, less

well-known retailers benefit more from a generous (loose) return policy than do well-known competitors (Jeng 2017). This also means that loose return policies (such as MBG) help low-quality retailers, not high-quality retailers (McWilliams 2012). An MBG policy may also be a strategic tool to help retailers develop their store brands (Huang and Feng 2020).

The implementation of return policies is also related to the decision-making regarding the channel structure, personalized pricing and discount pricing (i.e., Davis et al., 1995; Moorthy and Srinivasan 1995; Ferguson et al., 2006; McWilliams 2012; Akçay et al., 2013; Hsiao and Chen 2015; Xia et al., 2016; Chen and Chen 2017a; Chen and Chen 2017b; Li et al., 2018; Khouja et al., 2019). The return rate and return processing cost of the direct channel (online channel) are regarded as the key factors for whether a manufacturer (or retailer) implements a return policy and decides the channel structure (Xia et al., 2016). Most studies (i.e., Hsiao and Chen 2014; Chen and Chen 2017a; Chen and Chen 2017b) assert that if the net salvage value of the returned product is positive, the retailer should provide MBG. However, others (i.e., Li et al., 2018) assert that even if the net residual value of the returned products in the direct channel is negative, the manufacturer operating the direct channel can still provide MBG. It should also be noted that a return policy is not applicable to all online retailers and consumers; only when the reliability of the product is moderate and the value in the low-end segment of the market is moderate can online retailers provide returns (Hsiao and Chen 2015). Moreover, when consumers are overly sensitive to the return policy, providing a return policy may harm online demand and aggregate demand (Li and Jiang 2019).

Personalized pricing (PPS) will make online channels more attractive, making online retailers more likely to adopt an MBG return policy (Chen and Chen 2017b). Under certain conditions, the adoption of MBG and PPS by retailers in physical channels and the adoption of MBG by manufacturers in their direct channels may create a win-win situation. However, the adoption of PPS by manufacturers in their direct channels will result in both retailers and manufacturers failing (Li et al., 2018).

In addition, a recent study (Khouja et al., 2019) analyzed the impact of the duration of a return policy on retailers who offer regular price discounts and those who do not. The results revealed that for retailers that provide consumers with regular price discounts, it is optimal to provide a price adjustment policy that provides the same duration as the return policy, and it can greatly reduce the cost of the return policy. Furthermore, a recent study (Cao et al., 2020) focused on the return policy of remanufactured products. The authors showed that when the carbon tax price is high, retailers are more likely to choose not to provide online return services for remanufactured products.

The above literature shows that return policies have attracted the attention of researchers. With the popularization of return policies and the enforcement of laws (for example, China's "Consumer Rights Protection Law" (China 2017) stipulates that, except for special products, commodities purchased online are allowed to be returned by consumers without reason within seven days from the date of arrival), online retailers often provide consumers with more relaxed return services, such as MBG, to reduce consumers' purchase risks and to comply with the law. However, as mentioned above, online retailers are also under pressure from return losses. To resolve such contradictions, return insurance has been introduced.

Although the above literature discusses return policies from various perspectives, it has not yet considered how online retailers should implement pricing and how to choose a return policy under the premise of introducing return insurance. This paper attempts to fill this gap in the research by discussing the pricing strategy and the return policy and whether to offer return insurance for online retailers.

2.2. Internet return insurance

The second related stream of literature is on Internet return insurance strategy. When studying insurance, most researchers focus on traditional offline insurance (i.e., De and Florence 1945; John 1958; Hu and Scott, 2007; Sydnor 2009; Benartzi et al., 2011; Yang 2012; Barberis 2013; Barseghyan and Molinari 2013; Zhang and Hu, 2013; Geng et al., 2017). However, in this article, we discuss a new type of Internet insurance product: "return insurance".

Return insurance is similar to transport or cargo insurance, which is one of the oldest and most developed types of insurance. In traditional insurance models, the pricing of transport or cargo insurance is based on buyers, sellers, objects, shippers, and even seasons (De and Florence 1945; John 1958). In general, there have been two periods in the return insurance industry. Before the new Internet return insurance was developed, the insurance premium charged equaled 5% of the freight cost (Yang 2012). Support vector machines (SVMs) are often used to analyze the factors that affect the buyers and sellers of insurance (Zhang and Hu, 2013). When the new Internet return insurance was developed, the compensation for online consumers was set equal to the cost of reverse logistics (Geng et al., 2017).

However, there is little research on the issue of Internet return insurance. Marotta, Martinelli et al. (2017) summarize the basic understanding of Internet insurance available to date, discuss the unique factors of this type of insurance in contrast to general insurance, and use various empirical methods to analyze and compare the market factor responses of Internet insurance. However, their research did not involve Internet return insurance. With the popularity of e-commerce and online shopping, to resolve the dispute over who should bear the return freight, Internet return insurance was born. Internet return insurance (hereinafter referred to as "return insurance") is a way to deal with return risks and to achieve a generous and economically viable return policy for online products (Geng et al., 2017). Based on this, Geng et al. (2017) constructed a profit maximization model for online retailers and insurance providers based on selected market response parameters, especially the uncertainty of product fit, to solve for the rationality premium and return freight compensation of online products. However, the type of return insurance they discussed was only buyer's return insurance, that is, the type of return insurance purchased by the consumers themselves. Their research did not discuss the seller's return insurance, that is, the return insurance purchased by retailers for consumers and presented to consumers, which is the GR scenario discussed in this article. Following Geng et al. (2017), Lin et al. (2020) considered the situation where retailers provide consumers with return insurance. Their study found that whether return insurance brings benefits to retailers depends on the unit premium and the return shipping cost. However, none of the above-mentioned research discusses the two return insurances (seller's return insurance and buyer's return insurance) at the same time, and the consumer demand function they construct is linear and does not consider the use of the consumer utility function. In fact, the use of the utility function can better reflect the relationship between the introduction of return insurance and the consumer's purchasing decision. In addition, Fan and Chen (2020) studied when e-retailers should provide free return insurance. However, their research did not involve discussions about the pricing strategy of the insurance provider's return insurance and consumer utility surplus.

Compared with the existing research, we considered all possible stakeholders (the online retailer, the insurance provider, and the consumers) of return insurance in consumer returns and return insurance implementation modes (seller's return insurance and buyer's return insurance). Therefore, our research is more comprehensive. To this end, we selected some studies related to our work as a comparison to clearly illustrate the similarities and differences between the relevant literature and this article in Table 1.

In summary, our study differs from the previous works in three respects. First, our work differs from the research on traditional offline insurance because we consider a new type of Internet insurance product – "return insurance" – and discuss two types of return insurance: seller's return insurance and buyer's return insurance. Second, when studying return policy and pricing strategy, most researchers ignore the impact of the return insurance strategy on the return policy and pricing strategy of

Table 1
Comparison among recent relevant studies.

Literature	Utility function	Return policy	Buyer's return insurance	Seller's return insurance	Pricing of return insurance	Consumer surplus
Hsiao and Chen (2014), Chen and Chen (2017a), Chen and Chen (2017b), Alaei et al. (2020)	✓	✓	×	×	×	×
Li et al. (2018), Khouja et al. (2019), Cao et al. (2020), Huang and Feng (2020)	✓	✓	×	×	×	×
Marotta et al. (2017)	×	×	×	×	×	×
Geng et al. (2017)	×	✓	✓	×	×	×
Lin et al. (2020)	×	✓	×	✓	×	×
Fan and Chen (2020)	✓	✓	✓	✓	×	×
This article	✓	✓	✓	✓	✓	✓

online retailers. However, in practice, return insurance has been applied to online shopping, which not only helps online retailers to provide a more relaxed return policy but also helps consumers by reducing return costs. Therefore, we further study the relationship among the online retailer's pricing strategy, return policy and return insurance strategy. In addition, we also considered the pricing strategy of insurance providers' return insurance, and based on consumer surplus, we provided consumers' recommendations on whether to purchase return insurance. Third, previous investigations have not considered the impact of return insurance on consumer purchase decision-making. However, the introduction of return insurance has changed the conditions under which consumers decide to purchase, and a utility function can better reflect the relationship between the two. In the absence of return insurance, a consumer's purchase decision relates only to his or her valuation; in the presence of return insurance, a consumer's purchase relates not only to his or her valuation but also to the insurance premium and compensation fee of the return insurance. Therefore, this paper further considers the impact of this important factor on consumer purchase decision-making. Based on the above, this paper attempts to study the combination of return insurance, pricing strategy and return strategy by constructing mathematical models of four scenarios. Moreover, we consider consumer choice regarding return insurance based on consumer surplus. Our study attempts to fill the gap in the literature, and our results offer additional managerial implications.

3. Problem description and modeling

3.1. Problem description and assumptions

In this paper, we consider a sales system to consist of an insurance provider (represented by the subscript "I") and an online retailer (represented by the subscript "R"). The online retailer sells consumers a product in an online channel, and the insurance provider offers an "online shopping return freight loss insurance service" to the online retailer and consumers, which is called "return insurance" in this paper. The online retailer sells a product online at price p with a unit operating cost of c. The online retailer decides whether to provide an MBG return policy to consumers and whether to offer return insurance free of charge; consumers choose whether to purchase the product and whether to purchase return insurance when the online retailer does not offer it. If a consumer's utility is greater than or equal to zero, the consumer purchases the product; however, due to the lack of "touch and experience" in online shopping, consumer perceptions of products may be inaccurate. This also means that a consumer may be dissatisfied with a product after receiving it. If the consumer is dissatisfied, based on the online retailer's return policy (whether consumers are provided with an MBG return service and whether free return insurance is gifted) and the consumer's decision-making (whether return insurance was bought when the online retailer did not gift free return insurance), there are four possible cases:

(1) Case NN: The online retailer does not accept returns, and this case is indicated by the superscript "NN". Here, consumers cannot return products even if they are dissatisfied with them.
(2) Case GN: The online retailer offers an MBG return policy that allows consumers to return products with which they are dissatisfied but does not provide free return insurance to each consumer, and consumers do not purchase return insurance themselves. We use the superscript "GN" to indicate this case. In other words, in this case, although the online retailer offers an MBG return policy, consumers must bear a return freight cost, namely, the reverse freight, if they return products.
(3) Case GR: The online retailer offers an MBG return policy and provides each consumer free return insurance, and this case is indicated by the superscript "GR". In this case, the online retailer provides consumers with return insurance free of charge, which alleviates, to a certain extent, consumers' purchase uncertainty, and this may encourage consumers to purchase and increase their willingness to pay for products.
(4) Case GC: The online retailer offers an MBG return policy but not return insurance, and consumers purchase their own return insurance. This case is indicated by the superscript "GC". In this case, because the consumer purchases return insurance, if the consumer returns an unsatisfactory product, the return insurance will substantially reduce the return shipping cost that he or she faces.

It should be noted that in our model, the case NN is the benchmark for comparison. In practice, this situation also exists. For example, the online products sold in this situation are generally special products that are not subject to mandatory returns within the scope of the law, such as consumer-customized products and fresh and perishable products. In the cases of GN, GR and GC, for nonspecial products, due to legal constraints (i.e., "Interim measures for the return without reasons of commodities purchased online within seven days" (China 2017) in Chinese Consumer Rights Protection Law or "Consumers have 14 calendar days to change their minds and return goods bought at a distance or off-premises and receive a full refund" (EU 2014) in the EU Consumer Rights Directive), retailers must provide MBG return services within a certain period of time; for special products, retailers may consider whether to provide MBG return services. This forms the significance of our research, especially for special products or nonspecial products that exceed the legally prescribed return period for no reason, that is, whether online retailers should provide an MBG return policy and whether they must give consumers additional free return insurance. Of course, for some countries where dealers do not have obligations to accept returns, all products are applicable to the four cases.

In addition, since the GR and GC situations involve return insurance, to better explain its role, we use Internet return insurance jointly launched by the China Huatai Insurance Group and the Taobao platform to explain consumer purchases and returns (if necessary) and the compensation process, as shown in Fig. 1. For some products sold on the Taobao platform, online retailers provide MBG return services. When

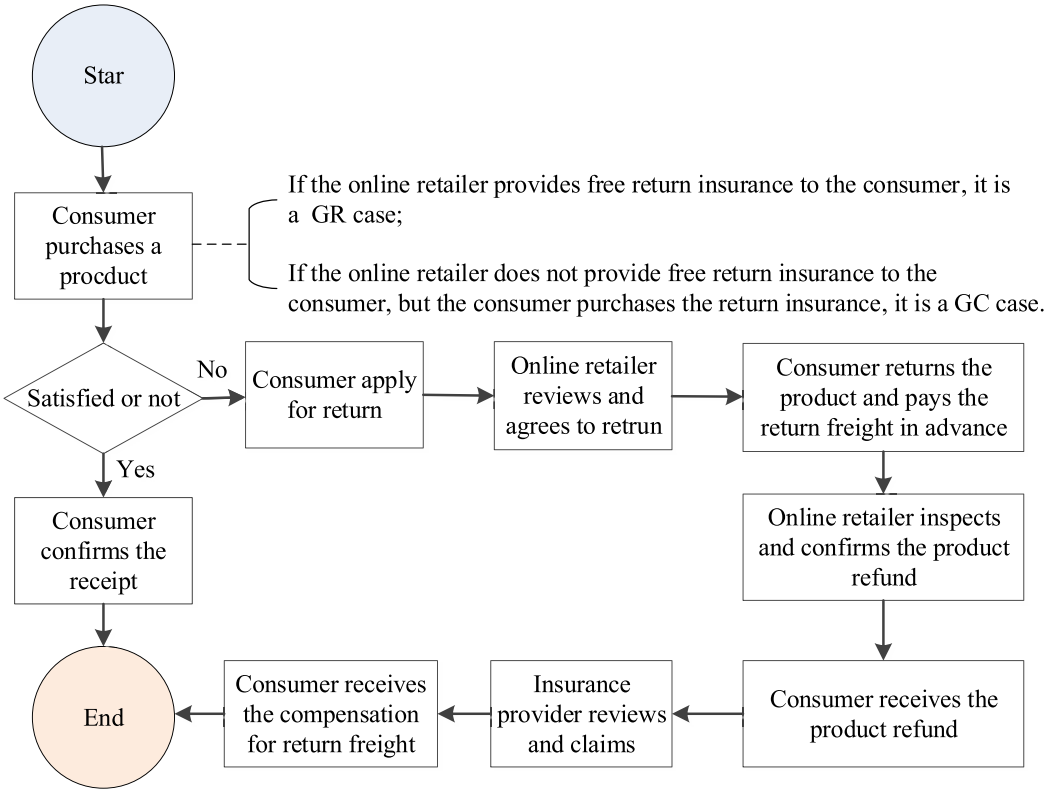


Fig. 1. Compensation process of Internet return insurance.

the online retailer provides free return insurance to consumers (GR case) or when consumers buy their own return insurance (GC case), if the consumer is satisfied with the product purchased online, the receipt will be confirmed, and the insurance agreement will be terminated. If the consumer is not satisfied with the product and needs to return it, he or she will apply to the online retailer for a return. For nonspecial products, the system will automatically review and pass the return application within the reasonable return period prescribed by law. For special products, the customer can return the product only after the online retailer accepts his or her return application. Then, the consumer must pay the return freight in advance and inform the online retailer of the logistics number of the delivery. The Taobao platform system can automatically file such claims. The insurance provider will review and process the claim within a certain period (usually within 72 h). When the review is passed, the customer will receive freight compensation, and the transaction process and the insurance agreement will be terminated. This also means that the role of return insurance is that when consumers initiate a return refund request, the insurance provider provides an insurance service that compensates for the reverse logistics freight caused by returns. However, it should be noted that the amount of return compensation that a consumer can receive is determined by the insurance provider and does not depend on the reverse logistics freight that the consumer actually pays for the return.

Furthermore, in this paper, we make the following assumptions.

(1) The potential market demand is 1.
(2) If the online retailer allows consumers to return a product, a full refund for the returned product will be provided after the return.
(3) Each consumer purchases at most one unit of the product without considering the loss of time involved in a product return.
(4) Consumers are rational and do not retain useless products, and there are no speculative returns.
(5) Because consumers cannot "touch and experience" an actual product bought via the online channel, they are uncertain about whether the product will meet their needs before they receive it. Therefore, we use the matching rate λ ($0 < \lambda < 1$) to indicate the possibility that a product purchased online will meet the consumer's needs, specifically, the consumer's satisfaction with the product.

3.2. Modeling

3.2.1. Case NN

As a benchmark for comparison, we first discuss case NN in which returns are not allowed. In the NN model, the online retailer does not offer an MBG return policy. Similar to Chen and Chen (2017b), Hsiao and Chen (2015) and McWilliams (2012), we assume that the consumer's willingness to pay for a product is v, and v is uniformly distributed on [0, 1]. If the consumer is satisfied with the product purchased online, i.e., it meets the consumer's needs, he or she obtains utility v-p. However, if the customer is not satisfied with the product, he or she obtains utility 0-p because the online retailer does not offer an MBG return policy. Therefore, in case NN, the expected utility of a customer is:

$$U_R^{NN} = \lambda(v-p) + (1-\lambda)(0-p) = \lambda v - p \tag{1}$$

Obviously, when $U_R{}^{NN} \geq 0$ or $v \geq v_1 = \frac{p}{\lambda}$, the consumer chooses to buy the product. Therefore, the product demand is:

$$D_R^{NN} = \int_{v_1}^{1} dv = 1 - v_1 = 1 - \frac{p}{\lambda} \tag{2}$$

The profit of the online retailer is:

$$\max \ \Pi_R^{NN}(p) = (p-c)D_R^{NN} \tag{3}$$

Proposition 1. *In case NN, the optimal sales price for an online retailer is* $p_R^{NN*} = \frac{c}{2} + \frac{\lambda}{2}$, *the online retailer's market share is* $D_R^{NN*} = \frac{\lambda - c}{2\lambda}$, *and the*

optimal profit for the online retailer is $\prod_R^{NN^*} = (c-\lambda)^2/4\lambda$.

The Proof of Proposition 1 is set forth in Appendix A.

Note that guaranteeing that $D_R^{NN^*} \geq 0$ requires the assumption that $\lambda \geq c$, which holds in the full text. Proposition 1 shows that the higher the unit operating cost and the matching rate are, the higher the sales price of the online retailer; the market share and profit of the online retailer also increase in the matching rate; however, the market share and profit of the online retailer decrease in the unit operating cost.

Next, we consider consumer surplus. In case NN, if the product meets the consumer's needs, the consumer surplus is $\lambda \int_{\hat{v}}^{1} (v-p)dv$, and if the product does not meet the consumer's needs, the consumer surplus is $(1-\lambda)\int_{\hat{v}}^{1}(0-p)dv$. In summary, we can obtain the total consumer surplus in case NN as $CS^{NN} = \lambda \int_{\hat{v}}^{1}(v-p)dv + (1-\lambda)\int_{\hat{v}}^{1}(0-p)dv$. By substituting $p_R{}^{NN^*}$ into the previous formula, we obtain the optimal total consumer surplus as $CS^{NN^*} = (c-\lambda)/8\lambda$.

Obviously, in case NN, the higher the matching rate is, the higher the consumer surplus, while the unit operating cost has a negative relationship with consumer surplus.

3.2.2. Case GN

In the GN model, the online retailer offers an MBG return policy; however, neither the consumer nor the online retailer buys return insurance. Then, if a consumer is satisfied with a product, he or she can obtain utility *v-p*; conversely, if the consumer is not satisfied with the product, he or she can obtain utility 0-*t*, where *t* is the reverse freight cost for the consumer's return. Therefore, in case GN, the expected utility of a customer is:

$$U_R^{GN} = \lambda(v-p) + (1-\lambda)(0-t) = \lambda(v-p) - (1-\lambda)t \tag{4}$$

Obviously, only when $U_R{}^{GN} \geq 0$ or $v \geq v_2 = p + \frac{1-\lambda}{\lambda}t$ does the consumer purchase online. Therefore, the product demand is:

$$D_R^{GN} = \int_{v_2}^{1} dv = 1 - v_2 = 1 - p - \frac{1-\lambda}{\lambda}t \tag{5}$$

We assume that the residual value of the returned product is *S*; *T* is the cost of (loss from) processing the returned product for the online retailer such as the cost of (loss from) repackaging or other handling of the returned product. Note that if the customer is not satisfied with the product, the reverse freight cost *t* incurred by the customer is only the shipping cost of returning the product through the online channel; the online retailer must bear not only the positive delivery cost but also the handling costs of the returned product, such as the cost of repackaging the returned product. Therefore, we assume that $T > t$, and this assumption is consistent with reality. Then, the profit of the online retailer is:

$$\max \ \Pi_R^{GN}(p) = [\lambda p - c + (1-\lambda)(S-T)]D_R^{GN} \tag{6}$$

Proposition 2. *In case GN, the optimal sales price of the online retailer is* $p_R^{GN^*} = (\lambda + c - (1-\lambda)E - 2(1-\lambda)t)/2\lambda$, *the online retailer's market share is* $D_R^{GN^*} = \Delta/2\lambda$, *and the optimal profit for the online retailer is* $\Pi_R^{GN^*} = \Delta^2/4\lambda$, *and here* $\Delta = (\lambda - c) + (1+\lambda)(S-T-t)$.

The Proof of Proposition 2 is set forth in Appendix A.

To facilitate analysis and interpretation, similar to Chen and Chen (2017b), we set $E = S - T - t$ as the net residual value of a returned product and $\Delta = (\lambda - c) + (1-\lambda)(S-T-t)$ as the average net expectation, in which $\lambda - c$ is the maximum net expected value ($v = 1$) when the consumer is satisfied, and $(1-\lambda)E$ is the net expected value when the consumer is not satisfied. Note that to ensure that $D_R^{GN^*} \geq 0$, we must assume that $\Delta \geq 0$, and this assumption holds throughout the text.

The Proof of Table 2 is set forth in Appendix A.

Table 2
The optimal sales price, market share, profit, and consumer surplus in case GN.

	$p_R^{GN^*}$	$D_R^{GN^*}$	$\Pi_R^{GN^*}$	CS^{GN^*}
λ	↓	↑	↑	↑
c	↑	↓	↓	↓
t	↓	↓	↓	↓
E	↓	↑	↑	↑
Δ	–	↑	↑	↑

According to Proposition 2 and Table 2, in case GN, the optimal price decreases in the matching rate, and the online retailer's market share and profit increase in the matching rate; however, they decrease in consumers' return freight cost. The price increases in unit operating costs; however, it decreases in the net residual value and the return freight cost. As the price decreases, the market share of the online retailer increases, which means that the online retailer alleviates the negative impact of consumer returns on its market share by reducing the selling price. In addition, as the net residual value of the returned product or the system's average net expectation Δ increases, the market share of the online retailer increases and its profit increases. This means that the online retailer should use appropriate means to reduce the cost of handling consumer returns and other approaches to improve the net residual value of returned products in the online channel to increase the corresponding market share and profit. This is also confirmed in Chen and Chen (2017b).

Similarly, next, we analyze the consumer surplus in case GN. If the product meets the consumer's needs, then the consumer surplus is $\lambda \int_{p+\frac{1-\lambda}{\lambda}t}^{1}(v-p)dv$, and if the product does not meet the consumer's needs, the consumer surplus is $(1-\lambda)\int_{p+\frac{1-\lambda}{\lambda}t}^{1}(0-t)dv$. Therefore, the total consumer surplus in case GN can be obtained as $CS^{GN} = \lambda \int_{p+\frac{1-\lambda}{\lambda}t}^{1}(v-p)dv + (1-\lambda)\int_{p+\frac{1-\lambda}{\lambda}t}^{1}(0-t)dv$. By substituting p^{GN^*} into the previous formula, we obtain the optimal total consumer surplus as $CS^{NH^*} = (c-\lambda)^2/8\lambda$.

Obviously, in case GN, the consumer surplus increases in the matching rate and decreases in the return freight cost, and the greater the net residual value of the returned product is, the greater the consumer surplus. This shows that the larger the return loss borne by the consumer is, the smaller the consumer surplus; however, the matching rate and the net residual value of the returned product can alleviate the negative effect of the return freight cost on consumer surplus.

3.2.3. Case GR

In this section, because it involves return insurance, we introduce a return insurance provider, and the corresponding decision order is as follows:

(1) The insurance provider first sets the premium price of return insurance c_i and maximizes its profit.
(2) Then, the online retailer decides the sale price *p* and whether to offer consumers free return insurance and maximizes its profit.
(3) The consumer finally decides whether to buy.

In the GR model, the online retailer offers an MBG return policy and provides free return insurance to each consumer. If a return occurs, the online retailer provides the consumer a product refund *p*, and the insurance provider reimburses the consumers for the return freight cost, namely, the return freight compensation t_i. If the consumer is satisfied with the product, he or she can obtain utility *v-p*. In contrast, if the consumer is not satisfied with the product and needs to return it, he or she can obtain utility $0 - t + t_i$, in which t_i is return freight compensation to the consumer provided by the insurance provider. Therefore, in case GR, the consumer's expected utility function is:

$$U_R^{GR} = \lambda(v-p) - (1-\lambda)(t-t_i) \tag{7}$$

Obviously, consumers choose to buy only when their expected utility satisfies $U_R^{GR} \geq 0$ or the consumer's willingness to pay is satisfied by $v \geq v_3 = p + \frac{(1-\lambda)(t-t_i)}{\lambda}$. Therefore, in this case, the product demand is:

$$D_R^{GR} = \int_{v_3}^{1} dv = 1 - v_3 = 1 - p - \frac{(1-\lambda)(t-t_i)}{\lambda} \tag{8}$$

The profit of the insurance provider is the insurance income minus the return freight compensation, which can be expressed as:

$$\max \ \Pi_I^{GR}(c_i) = [c_i - (1-\lambda)t_i]D_R^{GR} \tag{9}$$

Since the online retailer allows consumers to return products and offers them return insurance free of charge, the cost of return insurance c_i should be subtracted from the online retailer's profits. Then, the online retailer's profit function can be expressed as:

$$\max \ \Pi_R^{GR}(p) = [\lambda p - (c + c_i) + (1-\lambda)(S-T)]D_R^{GR} \tag{10}$$

We use backward induction to obtain Proposition 3.

Proposition 3. *In case GR, the optimal sales price for the online retailer is* $p_R^{GR*} = (c + 3\lambda - (1-\lambda)E - 4(1-\lambda)(t-t_i))/4\lambda$, *and the premium price charged by the insurance provider for return insurance is* $c_i^{GR*} = \Delta/2 + (1-\lambda)t_i$. *The market share* D_R^{GR*} *and optimal profit* Π_R^{GR*} *of the online retailer and the optimal profit of the insurance provider* Π_I^{GR*} *are detailed in* Table 3.

The Proof of Proposition 3 is set forth in Appendix A.

Proposition 3 and Table 4 illustrate that as the return freight compensation t_i increases, the premium price of the return insurance and the sales price of the online retailer increase. This means that because the online retailer provides free return insurance, as the premium price of the return insurance increases, its operating cost increases. Thus, the online retailer subsequently increases its sales price to prevent some consumers from purchasing products, thereby reducing the risk of returns. The premium price of the return insurance decreases in the consumer's return freight cost, which means that as the premium price of the return insurance increases, to reduce the risk of consumer returns and encourage more retailers to purchase return insurance for consumers, insurance providers reduce the premium price of return insurance. We found above that the retailer can benefit from increasing the net residual value of the returned product or the average net expectation. The online retailer should increase the net residual value of the returned product or the average net expectation in the online channel by reducing the cost of processing customer returns, which is conducive to increasing the market share and profitability of its channels. Other parameters have the same effect on the optimal value as in case GN.

We next analyze consumer surplus. In case GR, if the product meets the consumer's needs, then the consumer surplus is $\lambda \int_{v_3}^{1}(v-p)dv$, and if the product does not meet the consumer's needs, then the consumer surplus is $(1-\lambda)\int_{v_3}^{1}(0-t+t_i)dv$. Thus, the total consumer surplus in case GR is $CS^{GR} = \lambda \int_{v_3}^{1}(v-p)dv - (1-\lambda)\int_{v_3}^{1}(t-t_i)dv$. By substituting p_R^{GR*} into the previous formula, we can obtain $CS^{GR^*} = \Delta^2/32\lambda$.

Table 3
Optimal values in case GR.

Optimal values for the online retailer	Optimal values for the insurance provider
$p_R^{GR*} = (c + 3\lambda - (1-\lambda)E - 4(1-\lambda)(t-t_i))/4\lambda$	$c_i^{GR*} = \Delta/2 + (1-\lambda)t_i$
$D_R^{GR*} = \Delta/4\lambda$	$\Pi_I^{GR*} = \Delta^2/8\lambda$
$\Pi_R^{GR*} = \Delta^2/16\lambda$	

Table 4
The optimal sales price, market share, profit, return insurance price and consumer surplus in case GN.

	p_R^{GR*}	D_R^{GR*}	Π_R^{GR*}	CS^{GR*}	c_i^{GR*}
λ	↓	↑	↑	↑	↑
c	↑	↓	↓	↓	↓
t	↓	↓	↓	↓	↓
t_i	↑	–	–	–	↑
E	↓	↑	↑	↑	↑
Δ	–	↑	↑	↑	↑

From Table 4, when the matching rate increases, consumer surplus rises. In contrast, the unit operating cost and the consumer's return freight cost have a negative relationship with consumer surplus. Consumer surplus increases in the net residual value or the average net expectation. This indicates that the net residual value and the average net expectation have a positive relationship with consumer surplus.

3.2.4. Case GC

In the GC model, the online retailer offers an MBG policy but does not provide consumers the corresponding free return insurance; instead, consumers purchase their own return insurance. If a return occurs, the online retailer provides consumers a product refund, and the insurance provider reimburses consumers for the return freight cost. The corresponding decision order is as follows:

(1) The insurance provider decides the premium price c_i of the return insurance and maximizes its profit.
(2) The online retailer decides the sales price p and maximizes its profit.
(3) The consumer decides whether to purchase the product and whether to purchase the return insurance.

Furthermore, because consumers buy return insurance themselves, if a consumer is satisfied with the product, he or she will obtain utility $v - p - c_i$. In contrast, if the consumer is not satisfied with the product and needs to return it, he or she will obtain utility $0 - t - c_i + t_i$. Therefore, in case GC, the expected net utility of a consumer purchasing a product online is:

$$U_R^{GC} = \lambda(v-p) - (1-\lambda)(t-t_i) - c_i \tag{11}$$

It is easy to see that only when $U_R^{GC} \geq 0$ or $v \geq v_4 = p + \frac{(1-\lambda)(t-t_i)+c_i}{\lambda}$ will consumers purchase online, and thus, the product demand is:

$$D_R^{GC} = \int_{v_4}^{1} dv = 1 - v_4 = 1 - p - \frac{(1-\lambda)(t-t_i) + c_i}{\lambda} \tag{12}$$

Similarly, the profit of an insurance provider is the insurance income minus the return freight compensation, which can be expressed as:

$$\max \ \Pi_I^{GC}(c_i) = [c_i - (1-\lambda)t_i]D_R^{GC} \tag{13}$$

The online retailer's profit function is:

$$\max \ \Pi_R^{GC}(p) = [\lambda p - c + (1-\lambda)(S-T)]D_R^{GC} \tag{14}$$

Similarly, Proposition 4 can be obtained via backward induction.

Proposition 4. *In case GC, the optimal sales price of the online retailer is* $p_R^{GC*} = (\lambda + 3c - 3(1-\lambda)E - 4(1-\lambda)t)/4\lambda$, *and the optimal return insurance price of the insurance provider is* $c_i^{GC*} = \Delta/2 + (1-\lambda)t_i$. *The market share of the online retailer* D_R^{GC*}, *the online retailer's optimal profit* Π_R^{GC*} *and the optimal profit of the insurance provider* Π_I^{GC*} *are detailed in* Table 5.

The Proof of Proposition 4 is set forth in Appendix A.

From Proposition 4 and Table 6, we find that the effect of the

Table 5
Optimal values in case GC.

Optimal values for the online retailer	Optimal values for the insurance provider
$p_R^{GC^*} = (\lambda + 3c - 3(1-\lambda)E - 4(1-\lambda)t)/4\lambda$	$c_i^{GC^*} = \Delta/2 + (1-\lambda)t_i$
$D_R^{GC^*} = \Delta/4\lambda$	$\Pi_I^{GC^*} = \Delta^2/8\lambda$
$\Pi_R^{GC^*} = \Delta^2/16\lambda$	

Table 6
The optimal sales price, market share, profit, return insurance price and consumer surplus in case GC.

	$p_R^{GC^*}$	$D_R^{GC^*}$	$\Pi_R^{GC^*}$	CS^{GC^*}	$c_i^{GC^*}$
λ	↓	↑	↑	↑	↑
c	↑	↓	↓	↓	↓
t	↓	↓	↓	↓	↓
E	↓	↑	↑	↑	↑
Δ	–	↑	↑	↑	↑

parameters on the optimal value in case GC is the same as that in case GR.

The Proof of Table 6 is set forth in Appendix A.

In case GC, if a product meets a consumer's needs, then the consumer surplus is $\lambda \int_{v_4}^{1}(v - p - c_i)dv$, and if the product does not meet the consumer's needs, the consumer surplus is $(1-\lambda)\int_{v_4}^{1}(0 - t - c_i + t_i)dv$. Thus, the total consumer surplus in case GC is $CS^{GC} = \lambda \int_{v_4}^{1}(v - p - c_i)dv + (1-\lambda)\int_{v_4}^{1}(0 - t - c_i + t_i)dv$. Substituting $p_R^{GC^*}$ into the previous formula, we obtain $CN^{NH^*} = (c-\lambda)^2/8\lambda$.

In case GC, the change in consumer surplus from changes in the parameters is the same as that in case GR.

4. Strategy selection

4.1. Selection of return policy

To examine the online retailer's decision to offer an MBG policy that allows consumers to return unsatisfactory products, we analyze and compare the online retailer's sales price, market share, and profit and consumer surplus in cases NN and GN and obtain managerial insights.

Proposition 5. *The impact of the return policy on the optimal decision is expressed as follows*:

(1) *when* $0 < t \le c + \lambda - (S - T)$, *then* $p_R^{GN^*} \ge p_R^{NN^*}$; *otherwise*, $p_R^{GN^*} < p_R^{NN^*}$;

(2) *when* $E \ge 0$, *then* $D_R^{GN^*} \ge D_R^{NN^*}$, $\Pi_R^{GN^*} \ge \Pi_R^{NN^*}$ *and* $CS^{GN^*} \ge CS^{NN^*}$.

Proposition 5 shows that the MBG return policy affects the optimal pricing strategy of the online retailer. When the consumer's return freight cost is lower, that is, when $t \le c + \lambda - (S - T)$, then the sales price is higher when the online retailer offers an MBG return policy than when it does not. Conversely, when the consumer's return freight cost is higher, that is, when$t > c + \lambda - (S - T)$, then the online retailer offers a lower sales price when it offers an MBG return policy. Note that as long as the net residual value of the returned product is greater than or equal to zero, the market share and the profit of the online retailer and consumer surplus are higher when an MBG return policy is offered than when one is not. This also means that as long as the net residual value of the returned product is greater than or equal to zero, that is, $E \ge 0$, the online retailer should offer an MBG return policy. This demonstrates that the online retailer should increase the net residual value of the returned product in the online channel by reducing the cost of processing customer returns and ensuring that the net residual value is positive, which can benefit the online retailer in implementing an MBG return policy.

4.2. Selection of return insurance

Regarding the online retailer's decision of whether to offer free return insurance to each consumer and the consumer's decision of whether to purchase return insurance, we analyze and compare the online retailer's sales price, market share, and profit and consumer surplus in cases GN, GR, and GC and obtain further managerial insights.

Proposition 6. *When an online retailer provides an MBG policy for customer returns, the impact of the online retailer's offer of free return insurance on the optimal decision is expressed as follows*: *when* $\Delta \ge 0$, *then* $p_R^{GR^*} > p_R^{GN^*}$, $D_R^{GN^*} \ge D_R^{GR^*}$, $\Pi_R^{GN^*} \ge \Pi_R^{GR^*}$ *and* $CS_R^{GN^*} \ge CS_R^{GR^*}$.

Proposition 6 shows that when the average net expectation $\Delta \ge 0$ and the online retailer offers free return insurance (compared with when they do not, i.e., case GR versus case GN), then the online retailer offering free return insurance will increase the sales price, which will reduce the market share and profit of the online retailer. This also means that when the average net expectation $\Delta \ge 0$, the online retailer should offer an MBG return policy but not free return insurance. It is interesting to note that the online retailer offering return insurance does not increase its market share or profit or increase consumer surplus. This may be because the cost to the online retailer increases when it offers free return insurance, which necessitates an increased sales price to prevent some consumers with lower perceived value from buying and, thus, to avoid losses from processing returns. However, the increase in the sales price reduces the market share and profit of the online retailer.

Proposition 7. *When an online retailer allows returns, the impact of consumer purchases of return insurance on the optimal decision is expressed as follows*: *when* $\Delta \ge 0$, *then* $p_R^{GN^*} > p_R^{GC^*}$, $D_R^{GN^*} \ge D_R^{GC^*}$, $\Pi_R^{GN^*} \ge \Pi_R^{GC^*}$, *and* $CS^{GN^*} = CS^{GC^*}$.

Proposition 7 shows that when the average net expectation is $\Delta \ge 0$ and consumers buy return insurance themselves (compared with when they do not, i.e., case GC versus case GN), due to the reduction in consumer surplus, even if the online retailer lower the sale price, the market share will still decrease, and the online retailer's profit will be further reduced.

Proposition 8. *Case GR versus case GC*: (1) *when* $\Delta \ge 0$, $p_R^{GR^*} > p_R^{GC^*}$; (2) $D_R^{GR^*} = D_R^{GC^*}$, $\Pi_R^{GR^*} = \Pi_R^{GC^*}$ *and* $CS^{GC^*} = CS^{GR^*}$.

A very interesting aspect of Proposition 8 is that regardless of whether the online retailer offers free return insurance or consumers purchase it themselves, as long as they have return insurance, the market share and profit of the online retailer and consumer surplus will remain the same; that is, $D_R^{GR^*} = D_R^{GC^*}$, $\Pi_R^{GR^*} = \Pi_R^{GC^*}$, and $CS^{GR^*} = CS^{GC^*}$. However, if the online retailer provides return insurance, it will increase its operating costs and thus increase its sales price; that is, $p_R^{GR^*} > p_R^{GC^*}$. In addition, by observing the optimal price, we find that when switching from the GC case to the GR case, the expense of the return insurance presented by the online retailer is actually transferred to the optimal pricing of the GR case. The amount transferred even exceeds the expense of the return insurance because $\lambda(p_R^{GR^*} - p_R^{GC^*}) = c_i$, i.e., $p_R^{GR^*} - p_R^{GC^*} \ge c_i$. This also means that consumers indirectly bear the expense of the return insurance.

Proposition 9. *The optimal decision for the insurance provider is* $c_i^{GR^*} = c_i^{GR^*}$, $\Pi_I^{GR^*} = \prod_I^{GC^*}$.

From Proposition 9, we observe another interesting phenomenon: the optimal pricing strategy of an insurance provider is unrelated to

whether the policyholder is an online retailer or a consumer, and its maximum profit does not differ between the two cases, indicating that the insurance provider is neutral, which helps the insurance provider gain trust from policyholders.

To better visually verify the above conclusions, we set the parameters and perform numerical simulations. To ensure that the parameters are within a valid range, we set the parameter values $c = 0.3, S = 0.25$, $T = 0.05$, $t = 0.02$, and $t_i = 0.005$. We then compare the optimal sales price, market share, profit, and consumer surplus in the four scenarios, as shown in Figs. 2–5.

From Propositions 5-8 and Figs. 2–5, we obtain the following managerial insights.

Insight 1: Online retailers should offer an MBG return policy but not free return insurance.

Insight 2: Consumers should choose products covered by an MBG return policy; however, they need not consider whether online retailers provide free return insurance.

Insight 3: Insurance providers should be neutral, which helps insurance providers gain trust from policyholders.

Insight 1 shows that it is recommended that online retailers offer an MBG return policy but not provide free return insurance (Propositions 5 and 6). This is because when the online retailer offers an MBG return policy without providing free return insurance (i.e., in the case of GN), the profit of the online retailer is greatest. Conversely, if the online retailer chooses to provide free return insurance, it does not increase its market share or profit. The online retailer's gift of free return insurance to consumers increases the cost of sales and increases its selling price but ultimately does not increase consumer surplus.

Insight 2 shows that consumers should choose products that provide an MBG return service for purchase; however, there is no need to consider whether online retailers provide free return insurance (Propositions 5, 7 and 8). This is because the consumer surplus is greatest when consumers choose this approach. This result is very interesting and runs contrary to our intuitive assumption. By observing Fig. 5, we find that even if the online retailer provides free return insurance, it has not increased the consumer surplus. This is because consumers can learn about the overall product matching through online product customer reviews before purchasing. Then, even if the online retailer gives consumers free return insurance, such a relaxed return policy to please consumers still does not change consumers' expected valuation of the product. Furthermore, it can be seen from Proposition 8 that the return insurance given by the online retailer has actually been added to the product price, even exceeding the value of the return insurance itself ($p_R^{GR^*} - p_R^{GC^*} \geq c_i$). Therefore, consumers will not pay a higher product

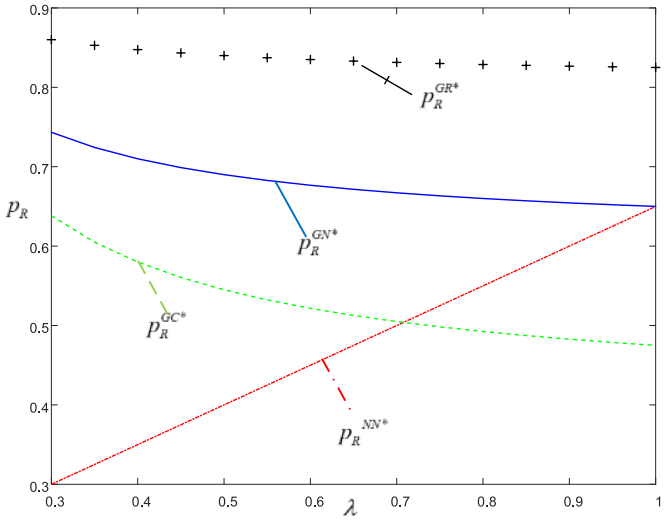


Fig. 2. Optimal sales prices of the online retailer.

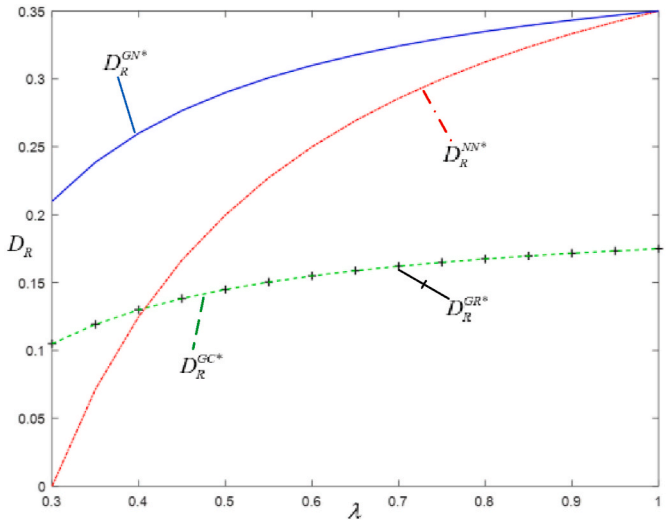


Fig. 3. Market shares of the online retailer.

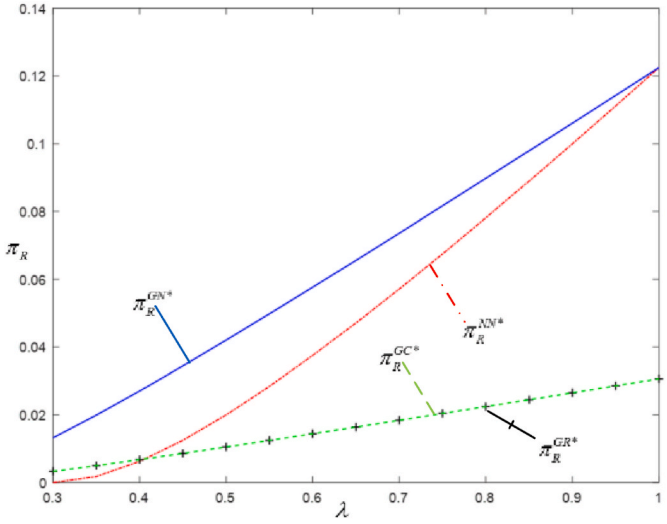


Fig. 4. Optimal profits of the online retailer.

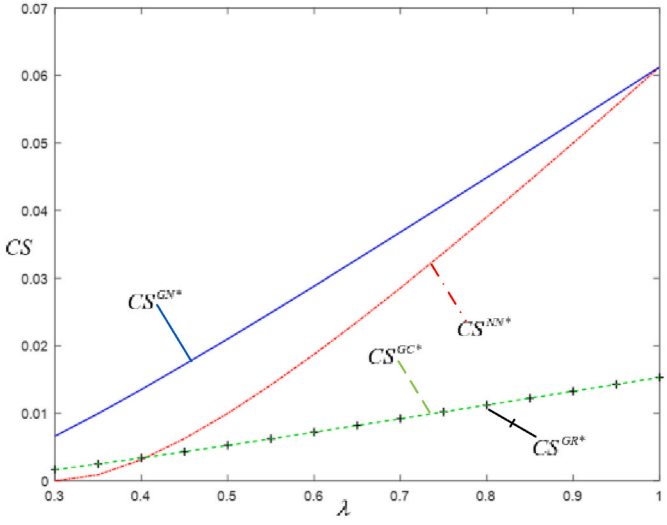


Fig. 5. Consumer surplus.

price to purchase a product with free return insurance. In general, the gift of free return insurance provided by online retailers will only lead to higher sales costs and more trouble for online retailers, causing their sales prices to rise while their market share and profits decline; if consumers consider choosing return insurance (either purchased by themselves or given by online retailers), it will not increase their average consumer utility surplus.

From Insight 3, for insurance providers, it is recommended that insurance providers be neutral (Proposition 9) because whether the policyholder is an online retailer or a consumer, the pricing and profit of insurance providers for return insurance will not change. This neutrality will help insurance providers gain trust from policyholders.

5. Conclusion

As online shopping continues to grow increasingly popular, the return policy of online retailers remains a topic of high academic interest. However, few researchers have studied the combination of Internet return insurance strategies and online retailers' pricing and return policies. To address the issue, we use a mathematical modeling method to construct models of four situations to fill the gap in the literature and provide corresponding managerial insights. The online retailer's optimal pricing and return strategy, the insurance provider's optimal insurance premium decision and consumers' optimal purchasing strategy are examined. Our study results not only provide theoretical support for the research of the mechanism of combining return insurance and return strategies but also guide practice.

We show the following: when the product's net salvage value is greater than or equal to zero, online retailers should offer an MBG return policy; however, there is no need for them to provide free return insurance because doing so does not increase their market share or profits. Moreover, it does not increase consumer surplus. This also explains the phenomenon whereby well-known brands online stores (such as Uniqlo (www.uniqlo.cn), Decathlon (www.decathlon.com) and JD (www.jd.com)) do not give consumers free return insurance but do provide MBG return service.

Consumers should choose products that provide an MBG return service for purchase; however, they do not need to consider whether online retailers give free return insurance. This also confirms the saying "There is no free lunch". In other words, consumers should not consider whether online retailers provide free return insurance. Nothing is free.

The optimal strategy for insurance providers does not differ depending on whether the policyholder is an online retailer or a consumer. An insurer should be neutral, which will help it to gain trust from policyholders. This result can be confirmed by the charging rules of return insurance. The charging rules of two types of return insurance sold on the Taobao platform are the same, regardless of whether it is the seller's version of return insurance or the buyer's version of return insurance. The charging rule is closely related to the return risk rate and has nothing to do with who is insured (Alipay 2020). This approach also reflects the fairness of return insurance and can help gain the trust of more insured persons. Therefore, in recent years, the number of insurance companies that provide return insurance on Taobao has grown from 1 in 2010 to 10 at present. In 2018, return insurance business accounted for nearly 50% of Internet property insurance. Compared with 2015, the compound growth rate of the premium income is as high as 84% (iResearch, 2019).

This paper has some limitations that should be mentioned. First, only one online retailer is considered, while in reality, multiple online retailers compete with each other. Second, we assume that the expected perceived value of the consumer when dissatisfied with the product is zero. However, in reality, consumers may communicate with online retailers when they are not satisfied with the product and then retain the product but ask for compensation. Therefore, future research could consider extensions that cover a more competitive environment, and the return and compensation strategies of the online retailer could be considered in greater detail. Finally, the main contribution of our work is the theoretical research on the mechanism of combining return insurance strategies and return policies through mathematical modeling, without in-depth and detailed empirical analyses. Therefore, empirical studies of the mechanism of combining return insurance strategies and return policies can also be considered as another further research direction.

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Declaration of Competing interest

The authors have no conflicts of interest to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2020.102350.

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